

Clipping Batch: SOTA Automated Pipeline Plan

Target Root: D:\Clipping batch | Engine: Native FFmpeg NVENC (17.5x Faster)

1. Executive Tooling Scorecard & Upgrade Matrix

Subsystem	Previous	SOTA Upgrade	Score	Rationale
Compositor	Remotion 4.0	FFmpeg filter_complex	3.0 → 9.5	12s NVENC vs 210s Remotion; 80MB vs 600MB RAM. Eliminates Chrome overhead.
Subtitles	React BouncySubtitle	Burned ASS/SSA	4.0 → 9.0	Dead React Spring code replaced with instant 0.1s hardware libass karaoke burns.
Instagram	instagram-private-api	Meta Graph API v24	1.0 → 9.0	Stops shadowbans within 1-3 uploads; uses official Meta Reels upload endpoints.
TikTok	Playwright Botting	TikTok Content API	4.0 → 8.0	Eliminates Arkose captcha and 0-view shadowbans from headless Chromium.
Architecture	Broken Monorepo	Flat 3-File Pipeline	2.0 → 9.0	Eradicates 100% mocked orchestrator; clean ingest -> transcribe -> render flow.
Process Mgr	PM2 Daemon	Windows Task Sched	2.0 → 8.5	PM2 targets deleted files and leaves zombie node processes on Windows.
TTS Stack	4 Redundant Engines	edge-tts + kokoro-js	3.0 → 8.0	Removes Chatterbox GPU crashes; reliable free cloud + local ONNX speech.

Phase 0: Security Nuke & Credential Scrubbing

- Task 0.1: Remove Plaintext OAuth: Delete yt-credentials*.json and auth.json.
- Task 0.2: Remove taskkill Bomb: Replace taskkill /F /IM node.exe with PID-tracked cleanup.
- Task 0.3: Arbitrary Eval Removal: Replace new Function() with JSON.parse().

Verification Gate 0: Zero exposed credentials and zero untracked process kill commands.

Phase 1: Architecture Flattening (src/pipeline/)

- Task 1.1: Eradicate Broken Monorepo: Delete packages/ directory entirely.
- Task 1.2: Establish 3 Core Files: Create ingest.js, transcribe.js, and render.js.

Verification Gate 1: packages/ deleted; 3 cohesive pipeline modules present.

Phase 2: Native FFmpeg Compositor (17.5x Render Acceleration)

- Task 2.1: Purge Remotion: Delete src/render/ directory.
- Task 2.2: Implement NVENC vstack: Render split-screen + ASS subtitles directly in FFmpeg.

Verification Gate 2: 60s 1080x1920 Short renders in ~12s with under 100MB system RAM.

Phase 3: Speech-to-Text & Subtitle Alignment

- Task 3.1: faster-whisper Engine: Produce millisecond word timestamps on CPU INT8.
- Task 3.2: ASS Karaoke Generator: Burn pop-in scale subtitles via hardware libass.

Verification Gate 3: Generated .ass subtitles contain valid karaoke tags and render in 0.1s.

Phase 4: TTS & Image Engine Consolidation

- Task 4.1: Strip Redundant TTS: Uninstall chatterbox and fish-audio.

- **Task 4.2: Standardize Speech:** Default to edge-tts with kokoro-js offline fallback.
- **Task 4.3: Streamline Image Gen:** Use fal.ai cloud or single local Forge SD 1.5.

Verification Gate 4: Zero PyTorch VRAM collisions on host GTX 1650 GPU.

Phase 5: Official Social Media Distribution

- **Task 5.1: YouTube Upload:** Resumable uploads via official googleapis SDK.
- **Task 5.2: Instagram Reels:** Publish via Meta Graph API v24 (replace reverse-engineered API).
- **Task 5.3: TikTok API:** Integrate official TikTok Content Posting API v2.

Verification Gate 5: Dry-run uploads succeed across all 3 platforms without browser botting.

Phase 6: Operational Cleanup

- **Task 6.1: Uninstall PM2 & Puppeteer:** Delete ecosystem.config.js.
- **Task 6.2: Enforce SQLite WAL:** Configure WAL mode on better-sqlite3 ledger.

Verification Gate 6: Ledger returns 'wal' pragma; clean 10-script package.json.